
Topology-aware adaptive variational quantum dynamics: growing McLachlan-optimal ansätze under qubit-connectivity constraints

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Abstract

Adaptive variational quantum dynamics (AVQDS) compresses unitary time evolution into shallow, problem-tailored circuits by growing a Pauli-rotation ansatz on the fly, keeping the McLachlan distance to the exact time-derivative manifold below a threshold. The growth rule, however, is connectivity-blind: two-qubit rotations are appended between arbitrary qubit pairs, and the cost of reconciling the resulting circuit with a device’s coupling graph is deferred to a separate compilation pass. We introduce *topology-aware AVQDS* (TA-AVQDS), which folds the connectivity constraint into the growth rule itself. Three components leave the underlying variational principle untouched: (i) an exactly optimized, wire-cost-minimal placement of the spin system onto the qubit graph; (ii) a distance-penalized operator-selection rule that adds a routing-cost surcharge $\eta [D(\pi(u), \pi(v)) - 1]$ to the trial McLachlan distance of each candidate two-qubit generator, with a variational safeguard that overrides the penalty whenever it would stall ansatz growth; and (iii) an exact compilation of the grown circuit onto the topology with greedy nearest-neighbor SWAP insertion, audited gate by gate. On eight qubits we simulate sudden quenches of the mixed-field Ising ring and finite-rate ramps of the Lieb–Schultz–Mattis chain on three coupling graphs: a 2×4 square lattice, an 8-qubit heavy-hex subgraph, and a linear chain. On the square lattice, which admits a Hamiltonian cycle, TA-AVQDS reproduces the unconstrained circuits exactly — zero routing overhead at identical fidelity. On heavy-hex, which admits no Hamiltonian path, it reduces the compiled CNOT count by 22–30% and the compiled depth by 13–26% relative to optimal placement plus post-hoc SWAP routing on three of four benchmarks, at final fidelities ≥ 0.996 ; on the fourth benchmark (LSM, $h_z = -0.7$) the penalty is counterproductive and we explain why. All gate counts, including every SWAP, are reported against the all-to-all CNOT convention, and every two-qubit gate is verified to lie on a coupling edge.

1 Introduction

Simulating the real-time dynamics of quantum many-body systems is both a central scientific task and one of the most natural applications of quantum computers [1, 2]. On near-term devices the standard approach — Trotter–Suzuki product formulas [3, 4, 5] — produces circuits whose depth grows linearly with simulation time and with the desired accuracy, quickly exhausting the error budgets of noisy hardware. Variational time-evolution methods replace the fixed product-formula circuit by a parametrized ansatz whose parameters evolve under a variational principle, typically McLachlan’s [6, 7, 8], and can reach the same accuracy with far fewer two-qubit gates [9, 10, 85, 87, 86, 88, 96]; the underlying classical variational principles have a long history in tensor-network dynamics [71, 69, 70]. The adaptive variant of this idea, AVQDS [11], goes further: the ansatz itself is grown during the simulation by trial-appending generators from an operator pool and keeping the one that minimizes the McLachlan distance, so the circuit structure adapts to the

state actually visited. At matched fidelity, the resulting circuits use up to two orders of magnitude fewer entangling gates than first-order Trotterization [11], and the framework has been extended to open systems, finite temperature, and imaginary-time evolution [13, 14, 15]. Related adaptive ideas have transformed static ground-state preparation in the ADAPT-VQE family of algorithms [19, 20, 21, 22], within the broader landscape of variational quantum algorithms [23, 24, 25, 26, 79, 80, 81, 82, 83, 84] and their classical optimization infrastructure [90, 89, 91, 92], whose ansatz expressibility and trainability limits are by now well characterized [93, 94, 95].

A structural assumption runs through all of these methods: the operator pool, and hence the generated circuit, treats all qubit pairs as equivalent. Hardware does not. Superconducting processors arrange qubits on heavy-hex [27, 28] or square [29, 30] lattices; trapped-ion and neutral-atom devices have their own, equally finite connectivities [31, 32, 78, 59]. A two-qubit gate between unconnected qubits must be synthesized from nearest-neighbor gates, conventionally by inserting SWAP networks, at three CNOTs per SWAP [72, 76, 77]. The standard workflow is therefore two-stage: an algorithm designs a circuit against a logical, all-to-all model, and a compiler — a stochastic or heuristic router [34, 35, 36, 37, 38, 39, 40, 61, 62, 75, 60, 74, 63] or an exact optimizer [41, 42, 43, 73] — repairs the connectivity mismatch afterwards. Compilation theory says this repair is expensive in general: routing on a sparse graph is NP-hard [44, 45], and worst-case overhead bounds are linear in the graph diameter [46]. Empirically, connectivity constraints measurably degrade variational algorithms [33] and have motivated connectivity-aware ansatz design [49, 50, 51, 52, 53, 54, 57, 58], most prominently fermionic swap networks [50].

It is natural to ask whether the two stages can be merged for *adaptive* dynamics: since the ansatz is built incrementally anyway, the connectivity cost of each candidate generator is known at the moment the growth decision is made, and the growth rule can simply price it in. This paper develops that idea. We keep the AVQDS framework — McLachlan equations of motion, Tikhonov-regularized inversion, adaptive time stepping, distance-thresholded growth — exactly as published, and modify only the *operator-selection objective*, supplementing it with an exactly optimized initial placement and an exact post-growth routing pass. The selection objective remains the McLachlan distance, to which we add a surcharge proportional to the topological distance the candidate’s support would have to be routed; a variational safeguard guarantees that the surcharge can reshuffle near-degenerate choices but can never stall growth. The method is stated completely in Sec. 3; a domain expert can reimplement it from that section alone.

Our findings, on four benchmark dynamics problems and three 8-qubit coupling graphs, are: (i) on the 2×4 square lattice, which admits a Hamiltonian cycle, topology-awareness is free — the compiled circuits are bit-identical to the unconstrained ones; (ii) on the heavy-hex subgraph, where routing is unavoidable, pricing connectivity into the growth rule reduces the compiled CNOT count by 22–30% and depth by 13–26% relative to the best placement plus shortest-path SWAP routing on three of four benchmarks, with final fidelities within 5×10^{-4} of the unconstrained baseline in both directions; (iii) the mechanism is the exploitation of near-degeneracies in the McLachlan landscape — the penalty redirects growth toward generators of equal variational power but lower routing cost, saturating at a small penalty strength η ; (iv) the method fails informatively on the LSM $h_z = -0.7$ ramp, where the degeneracy structure of the pool makes the penalty select a larger ansatz, and we analyze this failure in detail; and (v) the classical cost of the added logic is negligible — the trial-expansion loop dominates and is unchanged.

2 Background: adaptive variational quantum dynamics

We briefly restate the AVQDS framework [11], which we hold fixed throughout. The variational state is a pseudo-Trotter ansatz

$$|\Psi[\theta(t)]\rangle = \prod_{\mu=N_\theta}^1 e^{-i\theta_\mu(t)\hat{A}_\mu} |\Psi_0\rangle, \quad (1)$$

an ordered product of Pauli rotations with generators $\hat{A}_\mu$ drawn from an operator pool [72]. McLachlan’s variational principle [6], minimizing $\|(\partial_t + i\hat{\mathcal{H}})|\Psi\rangle\|^2$ over $\dot{\theta}$, yields the equations of motion

$$\sum_\nu M_{\mu\nu} \dot{\theta}_\nu = V_\mu, \quad M_{\mu\nu} = \text{Re} \left[\frac{\partial \langle \Psi |}{\partial \theta_\mu} \frac{\partial |\Psi\rangle}{\partial \theta_\nu} \right], \quad V_\mu = \text{Im} \left[\frac{\partial \langle \Psi |}{\partial \theta_\mu} \hat{\mathcal{H}} |\Psi\rangle \right], \quad (2)$$

integrated by forward Euler with an adaptively shortened time step such that $\max_\mu |\delta\theta_\mu| \leq \delta\theta_{\max}$. The inversion of M is stabilized by Tikhonov regularization, $M \rightarrow M + \xi \mathbb{I}$ with $\xi = 10^{-6}$. The accuracy of the ansatz at each time is measured by the squared McLachlan distance

$$L^2 = \left\| \left(\partial_t + i\hat{\mathcal{H}} \right) |\Psi[\theta]\rangle \right\|^2 = \text{Var}(\hat{\mathcal{H}}) - V^T M^{-1} V, \quad (3)$$

and whenever $L^2 \geq L_{\text{cut}}^2$ the ansatz is grown: each pool operator $\hat{A}_k$ is trial-appended (at $\theta = 0$, leaving the state invariant), its trial distance L_k^2 is evaluated through Eq. (3) using the border update of M and V , and the operator with the smallest L_k^2 is permanently appended; ties at $L_k^2 < L_{\text{cut}}^2$ are broken in favor of the smaller maximal parameter rate $|\dot{\theta}|_{\max}$. Growth repeats until $L^2 < L_{\text{cut}}^2$. With the Hamiltonian pool — the non-identity Pauli strings of the Hamiltonian — and parameters $L_{\text{cut}}^2 = 10^{-3}$, $\delta\theta_{\max} = 10^{-2}$, this procedure reproduces the reference dynamics of 8-qubit Ising and LSM chains at fidelities above 99.5% with $\sim 10^2$ two-qubit rotations [11].

Two features matter for what follows. First, the trial-distance landscape $\{L_k^2\}$ is structured but often *near-degenerate*: distinct pool operators can reduce L^2 by nearly identical amounts, because the Hamiltonian dynamics mixes pool generators into one another. Second, the growth rule acts purely on the basis of these distances — it has no knowledge of where each generator would physically act. It is precisely this slack that our method exploits.

3 Methodology: topology-aware ansatz growth

3.1 Problem statement

We are given a qubit coupling graph $G = (Q, E)$ with $|Q| = n$ vertices and edge set E ; a two-qubit gate is *compliant* iff it acts on an edge $\{q, q'\} \in E$. The simulated system is a spin chain whose sites $s = 0, \dots, n-1$ interact along its own interaction graph (a ring or a chain), generally different from G . A *placement* is a bijection $\pi : \{\text{sites}\} \rightarrow Q$; we write $D(q, q')$ for the graph distance (BFS shortest-path length) on G . The goal is a variational dynamics algorithm that (a) follows the AVQDS framework of Sec. 2 unchanged, (b) produces circuits in which every two-qubit gate is compliant, and (c) minimizes the compiled entangling cost, measured by CNOT count N_{cx} (2 CNOTs per two-qubit Pauli rotation, 3 per SWAP, all SWAPs counted) and by entangling depth (greedy layer packing of two-qubit gates; single-qubit rotations are absorbed into adjacent layers).

3.2 Overview

TA-AVQDS layers three components on the unchanged AVQDS core:

1. **Optimal static placement** (Sec. 3.3): one exact minimization, before the simulation, of the total wire cost of the Hamiltonian pool on G .
2. **Distance-penalized operator selection with a variational safeguard** (Sec. 3.4): the only change to the numerics of ansatz growth.
3. **Exact compilation with greedy SWAP routing and a compliance audit** (Sec. 3.5): applied to the finished circuit; unitarily exact in noiseless simulation.

Setting the single new hyperparameter $\eta = 0$ and using the identity placement recovers unconstrained AVQDS exactly.

3.3 Optimal static placement

Each two-qubit pool operator $\hat{A}_k$ acts on a pair of sites (u, v) ; let $w_{uv} = \sum_k: \text{supp}(\hat{A}_k) = \{u, v\} |c_k|$ be the total absolute Hamiltonian coefficient carried by the pair (for LSM, $w_{uv} = |J_x| + |J_y|$ evaluated at the ramp midpoint; for MFIM, $w_{uv} = |J|$). We solve

$$\pi^* = \arg \min_{\pi \in S_n} \sum_{(u,v)} w_{uv} D(\pi(u), \pi(v)) \quad (4)$$

exactly by exhaustive enumeration of all $n!$ placements with branch-and-bound ($n = 8$: 40320 evaluations, $O(1)$ s). All optima are enumerated; ties are broken deterministically by minimizing

the maximum bond distance $\max_{(u,v)} D(\pi(u), \pi(v))$, then lexicographically. This is the minimum-linear-arrangement problem on a graph metric [55, 56], tractable here because n is small; the same weighted wire-cost objective underlies placement passes in production compilers [36, 48].

The placement determines the *unavoidable* routing cost. If G admits a Hamiltonian cycle (Topology A below), π^* places every pool bond on an edge: $D(\pi^*(u), \pi^*(v)) = 1$ for all (u, v) , and no routing is ever needed. If G has no Hamiltonian path (Topology B), at least one bond sits at distance ≥ 2 under every placement, and SWAPs are unavoidable — the optimal placement minimizes, but cannot eliminate, this exposure.

3.4 Distance-penalized operator selection

During ansatz growth, AVQDS evaluates for every pool candidate $\hat{A}_k$ the trial McLachlan distance $L_k^2 = \text{Var}(\hat{\mathcal{H}}) - V_k^T M_k^{-1} V_k$ of the trial-extended ansatz. We modify the selection objective — and nothing else — to

$$\tilde{L}_k^2 = L_k^2 + \eta [D(\pi(u_k), \pi(v_k)) - 1] \quad \text{for two-qubit candidates on sites } (u_k, v_k), \quad (5)$$

and $\tilde{L}_k^2 = L_k^2$ for single-qubit candidates. The selection rule is otherwise the authors': append the candidate with the smallest $\tilde{L}_k^2$; if the winner already satisfies $\tilde{L}_k^2 < L_{\text{cut}}^2$, prefer among such candidates the one with smaller $|\dot{\theta}|_{\text{max}}$. The surcharge $\eta [D - 1]$ is exactly the per-rotation routing-cost proxy of the candidate: a rotation on a bond at distance D requires $D - 1$ additional nearest-neighbor SWAPs to execute under any path-based router. The penalty therefore prices into the growth decision what the compiler will later charge, in units of McLachlan distance squared; η converts between the two currencies.

Variational safeguard. A fixed surcharge could in principle override the variational principle itself: a candidate could win on $\tilde{L}^2$ while its *true* trial distance remains above threshold, with a strictly better true-distance candidate available — in effect, stalling growth to save routing cost. We forbid this: if the selected candidate has $L_{\text{sel}}^2 > L_{\text{cut}}^2$ while another candidate has $L_k^2 < L_{\text{sel}}^2 - \varepsilon$ (with $\varepsilon = 10^{-8}$), the selection falls back to the best true-distance candidate for that growth step. The penalty is thereby confined to its intended role: reshuffling *near-degenerate* selections. It can never increase the final McLachlan distance beyond what unconstrained growth would tolerate, and the accuracy of the ansatz remains governed by L_{cut}^2 alone.

Choice of η . The penalty only matters where trial distances are near-degenerate at the scale $\eta [D - 1]$; once η exceeds the typical near-degeneracy width, all topology-preferred reshuffling that is possible in principle has happened, and larger η changes nothing. We scan $\eta \in \{0, 0.02, 0.05, 0.1, 0.2, 0.4\}$ in Sec. 5.3 and find the compiled cost is exactly constant for all $\eta \geq 0.02$ on every benchmark; we fix $\eta = 0.1$, deep in the saturated regime, for all headline runs. The saturated circuit is independent of η over an order of magnitude, so the method introduces no tunable sensitivity.

3.5 Compilation, routing, and audit

The grown ansatz is an ordered list of logical Pauli rotations $e^{-i\theta_\mu \hat{A}_\mu}$ on spin sites. Compilation maps it onto G starting from π^* : a current site-to-qubit map σ is maintained; a two-qubit rotation on sites (u, v) whose qubits $\sigma(u), \sigma(v)$ are not adjacent is preceded by nearest-neighbor SWAPs that carry $\sigma(u)$ along a BFS shortest path to adjacency with $\sigma(v)$ (greedy routing; SWAPs are exact unitaries in noiseless simulation, so the routed circuit equals the logical circuit up to a tracked final qubit permutation, which is folded into observable evaluation). Single Pauli rotations decompose into 2 CNOTs plus single-qubit gates; each SWAP into 3 CNOTs. Every emitted two-qubit gate is checked against E and logged with its qubit pair and a compliance flag (per-step audit trail, reported in Sec. 5). The same router, applied to the circuit grown *without* the penalty, defines the smart-placement naive baseline of Sec. 4; the variant and the baseline therefore differ only in how the ansatz was grown, not in how it was compiled.

3.6 Positioning against prior work

Quantum compiling as a variational learning problem has been explored in [98, 99, 97]; our compilation pass is instead exact and classical, and the learning happens in the growth rule.

The components of TA-AVQDS are deliberately standard: wire-cost placement [55, 36, 48], shortest-path SWAP routing [34, 38], and exact subgraph/embedding optimization [41, 42, 43] are compilation staples, and routing costs enter reinforcement-learning and heuristic transpiler passes [47, 48]. The contribution is not a new router but the *coupling of routing cost to a variational growth rule*: the penalty (5) acts *before* the circuit exists, when an equivalent but cheaper generator can still be chosen. This is complementary to post-hoc routing [35, 34, 37], to connectivity-aware static ansatz design (hardware-efficient [33], fermionic swap networks [50], and topology-matched correlated ansätze [52, 53, 54]), and to connectivity-restricted adaptive growth for ground states [20, 22]: those methods restrict *what* may be added, whereas TA-AVQDS keeps the full Hamiltonian pool and reweights *when* each generator is preferred, with a safeguard tying the reweighting to the variational principle. The fermionic swap network of Ref. [50] interleaves routing with evolution in a fixed pattern dictated by fermionic locality; here the routing pattern is an emergent consequence of growth decisions made by the McLachlan principle itself. To our knowledge, pricing topological distance into the trial-distance selection of an adaptive *dynamics* algorithm has not been reported.

4 Experimental setup

Systems. Two 8-qubit benchmarks from the nonequilibrium physics of spin chains [64, 65, 68, 67, 66, 100], following Ref. [11]. (1) The mixed-field Ising model (MFIM) on a periodic ring, $\hat{H}_{\text{MFIM}} = -\sum_i Z_i Z_{i+1} + h_x \sum_i X_i + h_z \sum_i Z_i$ with $J = 1$, $h_x = -2$, $h_z \in \{0, 0.5\}$, suddenly quenched at $t = 0$ from $|0\rangle^{\otimes 8}$ and evolved to $T = 3$. The Hamiltonian pool has 16 operators ($h_z = 0$: $\{X_i\}$, $\{Z_i Z_{i+1}\}$ including the periodic bond $Z_7 Z_0$) or 24 ($h_z = 0.5$: adding $\{Z_i\}$). (2) The Lieb–Schultz–Mattis (LSM) chain with open boundaries, $\hat{H}_{\text{LSM}}(t) = -\sum_i [(1 + \gamma(t)) X_i X_{i+1} + (1 - \gamma(t)) Y_i Y_{i+1}] + h_z \sum_i Z_i$, with γ ramped linearly from 1 to -1 over $T = 3$ and then held at -1 for another T (total $2T = 6$), starting from the exact-diagonalization ground state of $\hat{H}(0)$, for $h_z \in \{-0.7, 1.6\}$. The pool is $\{Z_i\} \cup \{X_i X_{i+1}, Y_i Y_{i+1}\}$ (22 operators). Exact reference trajectories are computed by matrix exponentiation on a converged mesh ($\delta t = 2 \times 10^{-3}$ midpoint propagation for the time-dependent LSM; fidelity converged to $< 10^{-6}$ against a 5×10^{-4} mesh).

Coupling graphs. Topology A: the 2×4 square lattice (10 edges), which admits the Hamiltonian cycle $0-1-2-3-7-6-5-4-0$. Topology B: an 8-qubit heavy-hex subgraph (7 edges; four degree-1 vertices 0, 2, 4, 7), which admits no Hamiltonian path — a Hamiltonian path has at most two degree-1 endpoints — so pigeonhole forces at least one ring bond off-topology under every placement. Topology C (generalization test): the linear chain (7 edges), which admits a Hamiltonian path but no cycle. The graphs and the MFIM placement on B are drawn in Fig. 1.

Methods compared. (i) *Unconstrained AVQDS* [11], with the published hyperparameters $L_{\text{cut}}^2 = 10^{-3}$, $\xi = 10^{-6}$, $\delta\theta_{\text{max}} = 10^{-2}$, $\delta t_{\text{max}} = 0.05$, $\text{pthcut} = 7000$; CNOTs counted at 2 per two-qubit rotation (all-to-all convention). (ii) *Smart-placement naive routing*: the unconstrained circuit compiled with the optimal placement (4) and greedy shortest-path SWAP insertion (Sec. 3.5) — the strongest post-hoc baseline, since the placement is exact. (iii) *TA-AVQDS* with $\eta = 0.1$. All three use identical initial states, pools, and hyperparameters; growth decisions are based only on McLachlan distances, never on fidelity to the exact state. The variational engine is a bit-exact reimplement of the authors’ published code [12] on a fast Pauli-operator backend ($\mathcal{O}(2^n)$ per rotation instead of dense 2^{2n} matrices), verified to reproduce the authors’ trajectories and gate counts exactly; the topology-aware logic is confined to a selection-objective map and a compilation pass, with $\eta = 0$ recovering the baseline.

Metrics. Per time step: fidelity $f(t) = |\langle \Psi_{\text{exact}}(t) | \Psi_{\text{var}}(t) \rangle|^2$, cumulative N_{cx} , entangling depth, N_θ ; for MFIM the Loschmidt echo $|\langle \Psi_0 | \Psi(t) \rangle|^2$; for LSM the energy $\langle \hat{H}(t) \rangle$ and spin correlations $\langle X_0 X_1 \rangle$, $\langle X_0 X_{N-1} \rangle$, $\langle Y_0 Y_1 \rangle$, $\langle Y_0 Y_{N-1} \rangle$; per-gate topology-compliance audits; wall-clock seconds. Every run is deterministic (no sampling noise); all runs are reported.

5 Results

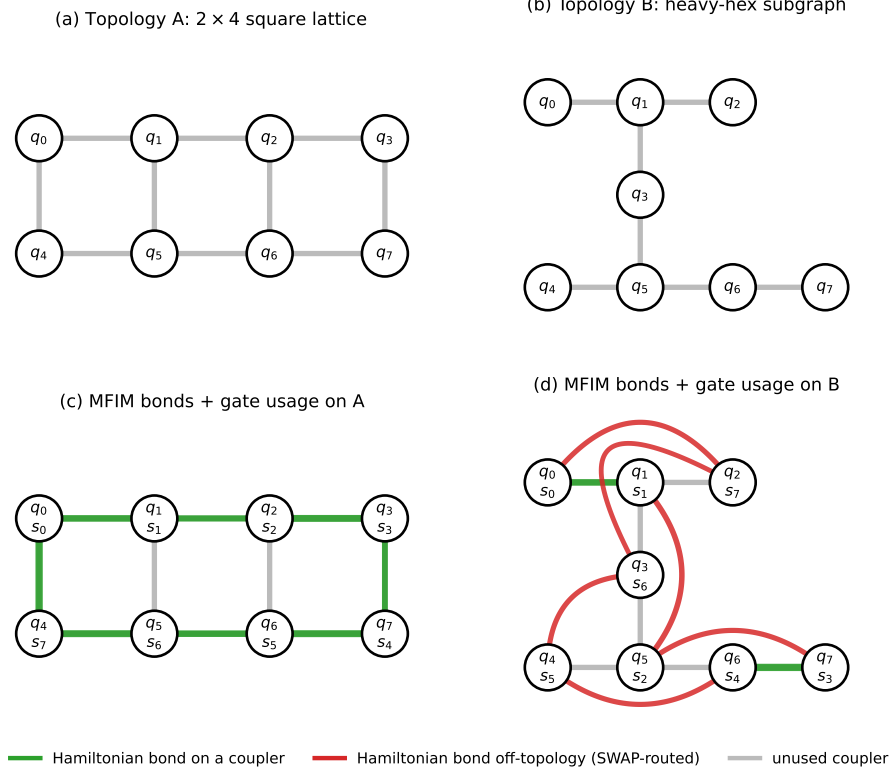


Figure 1: (a) Topology A, the 2×4 square lattice; (b) Topology B, the 8-qubit heavy-hex subgraph. (c,d) The MFIM interaction graph (a ring) placed on each topology by the optimal placement π^* , overlaid with the two-qubit-gate usage of the grown TA-AVQDS circuit (MFIM $h_z = 0.5$): straight green lines are ring bonds that land on couplers, curved red arcs are off-topology bonds that must be SWAP-routed, with line width proportional to usage count; nodes are labelled by physical qubit q_i and by the spin site s_j placed on it. On A (c) all eight bonds are couplers and no SWAP is ever emitted; no placement of the ring on B (d) puts all eight bonds on couplers.

5.1 Accuracy and gate counts

Tables 1 and 2 report the headline numbers for MFIM and LSM respectively; Table 3 collects the overhead multipliers, and Fig. 2 shows the full fidelity trajectories. Three statements summarize the data.

Topology A is free. On the square lattice, the optimal placement maps every pool bond onto a coupler, the penalty (5) vanishes identically, and both topology-constrained methods reproduce the unconstrained circuit exactly: identical fidelities, gate counts, and depths (multiplier 1.00 everywhere, zero SWAPs). The compliance is not approximate — the circuits are the same circuits, merely with qubit labels permuted along the Hamiltonian cycle.

On Topology B, growth-time pricing beats post-hoc routing on three of four benchmarks. Relative to the smart-placement naive baseline, TA-AVQDS reduces the compiled CNOT count by 27.6% (MFIM $h_z = 0$), 30.2% (MFIM $h_z = 0.5$), and 27.7% (LSM $h_z = 1.6$), and the compiled depth by 21.9%, 22.6%, and 26.9%. Against the all-to-all reference, the remaining overhead is a factor 1.93–2.42 in CNOTs where the naive baseline pays 2.67–3.46. Final fidelities stay within 5×10^{-4} of the unconstrained baseline in either direction on all four benchmarks — 0.9961 vs. 0.9957, 0.9973 vs. 0.9973, 0.9990 vs. 0.9991, and 0.9977 vs. 0.9976 — comfortably inside any 1% accuracy-preservation criterion, and in three cases the constrained run is marginally *more* accurate, because the redirected growth happens to add variational parameters (N_θ grows from 124 to 133 on MFIM $h_z = 0$ on topology C, for instance).

Table 1: MFIM sudden-quench benchmarks at $T = 3$: final-time fidelity, total CNOT count N_{cx} (2 per two-qubit rotation, 3 per SWAP, all SWAPs included), and entangling depth. “Naive routing” is the smart-placement baseline; TA-AVQDS uses $\eta = 0.1$. On Topology A both constrained methods coincide with the unconstrained circuit.

Method	Topology	MFIM $h_z = 0$			MFIM $h_z = 0.5$		
		fidelity	N_{cx}	depth	fidelity	N_{cx}	depth
Unconstrained AVQDS	— (all-to-all)	0.99566	134	25	0.99734	212	43
Naive routing	A	0.99566	134	25	0.99734	212	43
Naive routing	B	0.99566	377	96	0.99734	734	190
TA-AVQDS (this work)	A	0.99566	134	25	0.99734	212	43
TA-AVQDS (this work)	B	0.99610	273	75	0.99729	512	147

Table 2: LSM ramp-quench benchmarks at total time $2T = 6$; columns as in Table 1. The $h_z = -0.7$ case on Topology B is the failure case analyzed in Sec. 6.2.

Method	Topology	LSM $h_z = -0.7$			LSM $h_z = 1.6$		
		fidelity	N_{cx}	depth	fidelity	N_{cx}	depth
Unconstrained AVQDS	— (all-to-all)	0.99911	106	24	0.99758	104	24
Naive routing	A	0.99911	106	24	0.99758	104	24
Naive routing	B	0.99911	214	59	0.99758	278	78
TA-AVQDS (this work)	A	0.99911	106	24	0.99758	104	24
TA-AVQDS (this work)	B	0.99902	240	62	0.99769	201	57

The exception is LSM $h_z = -0.7$ on Topology B, where TA-AVQDS produces a *larger* circuit than naive routing (240 vs. 214 CNOTs) at equal fidelity. We analyze this failure in Sec. 6.2.

Topology C confirms the mechanism. On the linear chain, the LSM open chain embeds perfectly (multiplier 1.00), while the MFIM ring loses its periodic bond; TA-AVQDS reduces the compiled CNOT count by 34.9% ($h_z = 0$) and 37.8% ($h_z = 0.5$) relative to naive routing, with fidelities 0.9965 and 0.9973. The method’s advantage grows with the routing exposure of the benchmark on the graph, exactly as designed.

Compliance. Across all 24 constrained runs (4 benchmarks $\times$ 3 topologies $\times$ 2 methods), the per-gate audit records zero two-qubit gates outside the respective edge sets; every SWAP itself lies on an edge.

5.2 Physical observables

Gate counts are only meaningful at fixed physical accuracy. Figure 6 shows the Loschmidt echo for the two MFIM quenches: the TA-AVQDS circuit on Topology B tracks the exact echo through the collapse-and-revival structure, indistinguishably from the unconstrained circuit. Figure 7 shows the LSM energy and spin correlations along the ramp and the post-ramp hold; the constrained circuits reproduce the exact observables, including the symmetry breaking of the finite-rate ramp, with deviations invisible at the plotted scale.

5.3 Ablations

Penalty strength η . Figure 8 shows the compiled CNOT multiplier on Topology B as a function of η . The transition is sharp: $\eta = 0$ (no pricing) yields the naive-baseline circuits, and every $\eta \geq 0.02$ yields the identical saturated circuit on all four benchmarks. The method therefore has no meaningful hyperparameter sensitivity — any η above the near-degeneracy scale of the trial-distance landscape gives the same answer. We also ran the full η scan on Topology A, where the penalty vanishes by construction and all curves coincide with the unconstrained baseline, as required.

Pool restriction. A natural alternative to pricing distance is to *restrict* the pool to topology-aligned generators. On Topology A this is a no-op (all pool bonds are already on edges), and on Topology B the restricted pool is too poor to complete the simulation: growth stalls with “dynamic ansatz cannot further improve” on all four benchmarks, because the pool generators not on edges are precisely the

Table 3: Overhead multipliers relative to unconstrained AVQDS (CNOT-count and depth ratios) for the smart-placement naive baseline and for TA-AVQDS, on every (benchmark $\times$ topology) combination. Topology C (linear chain) is the third-topology generalization test. Values below 1.00 do not occur; bold marks the better constrained method on each row.

Benchmark	Topology A		Topology B		Topology C	
	naive	TA-AVQDS	naive	TA-AVQDS	naive	TA-AVQDS
MFIM $h_z = 0$	1.00 / 1.00	1.00 / 1.00	2.81 / 3.84	2.04 / 3.00	3.04 / 3.52	1.98 / 2.28
MFIM $h_z = 0.5$	1.00 / 1.00	1.00 / 1.00	3.46 / 4.42	2.42 / 3.42	3.49 / 2.98	2.17 / 2.21
LSM $h_z = -0.7$	1.00 / 1.00	1.00 / 1.00	2.02 / 2.46	2.26 / 2.58	1.00 / 1.00	1.00 / 1.00
LSM $h_z = 1.6$	1.00 / 1.00	1.00 / 1.00	2.67 / 3.25	1.93 / 2.38	1.00 / 1.00	1.00 / 1.00

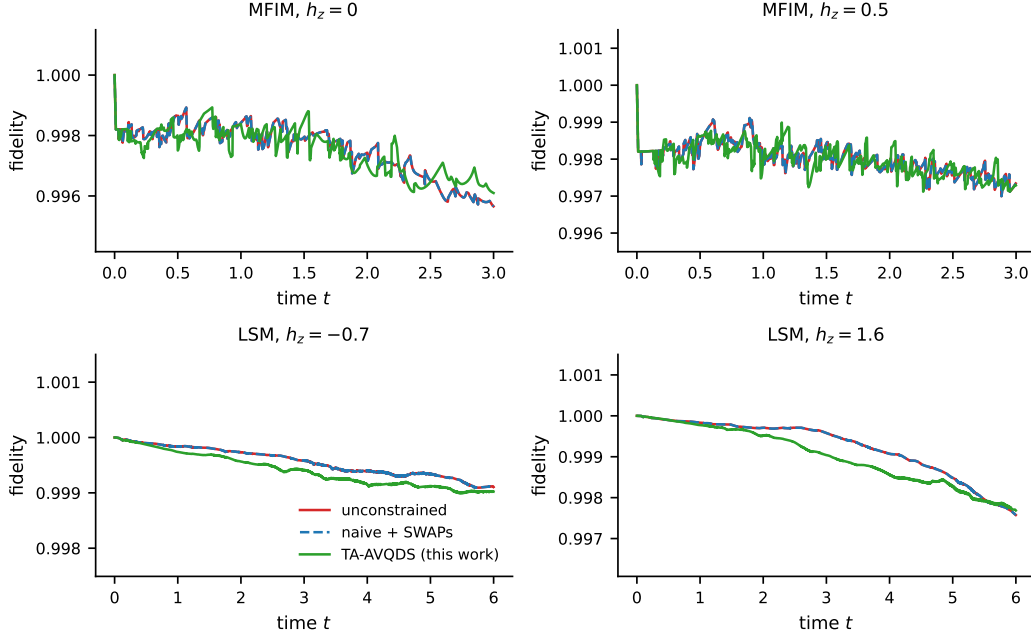


Figure 2: State fidelity versus time on Topology B (heavy-hex) for the four benchmarks, comparing unconstrained AVQDS (all-to-all reference), smart-placement naive routing, and TA-AVQDS ($\eta = 0.1$). The three methods remain within $\sim 4 \times 10^{-4}$ of each other at all times; the final values are listed in Tables 1 and 2. The corresponding Topology A curves are shown in Fig. 3; there the three methods coincide exactly.

ones the dynamics needs to rotate into. This negative result (logged in the attempts record) motivates the reweighting design of Eq. (5) over hard restriction.

6 Analysis

6.1 Mechanism: where the savings come from

The overhead of naive routing has two distinct sources, and TA-AVQDS removes only one of them. The *placement floor* is set by the graph: on Topology B, even the optimal placement leaves some Hamiltonian bonds at distance ≥ 2 (for MFIM, three bonds under the tie-broken optimum; the ring’s eight bonds cannot all fit on seven edges arranged without a Hamiltonian path). Rotations on those bonds need SWAPs no matter how the ansatz is grown — this cost is irreducible. The *selection exposure* is everything above the floor: the unconstrained growth rule chooses among near-degenerate candidates without regard to routing cost, and each choice of an off-topology generator adds a recurring $3(D - 1)$ -CNOT surcharge every time that generator reappears in the ansatz (pool operators are appended repeatedly, dozens of times).

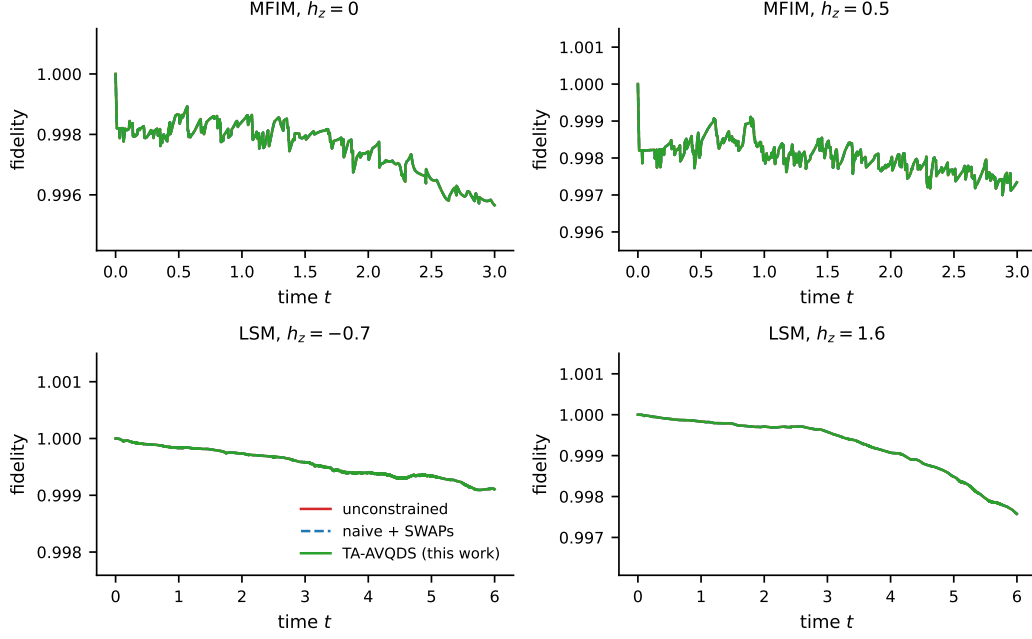


Figure 3: State fidelity versus time on Topology A (2×4 square) for the four benchmarks. On this topology the Hamiltonian-cycle embedding makes every pool bond a coupler, so the naive-routing and TA-AVQDS circuits coincide with the unconstrained circuit and all three curves lie exactly on top of each other (multiplier 1.00, zero SWAPs) — topology compliance is achieved at zero cost in both gate count and fidelity.

The penalty (5) converts the selection exposure into a growth-time decision. Two observations from the logs explain the magnitude of the effect. First, near-degeneracies are frequent: at many growth steps the best on-topology candidate and the best overall candidate reduce L^2 by amounts differing by less than $\eta(D-1) = 0.1$, so the penalty flips the choice; the safeguard fires regularly in the saturated regime (30–54 times per MFIM run, 8–9 per LSM run on Topology B), each time confining the penalty to a genuinely degenerate reshuffle, and the final fidelities move by at most 4×10^{-4} from baseline. Second, the flips compound: choosing an on-topology generator changes the state trajectory of the ansatz, which changes subsequent near-degeneracies, and the saturated circuit ends up with both fewer off-topology rotations *and* fewer SWAPs per remaining off-topology rotation — on MFIM $h_z = 0.5$, SWAP count drops from 174 to 116 while two-qubit rotations drop from 106 to 82. The sharp saturation in Fig. 8 is the signature that the penalty exhausts the available near-degenerate slack; beyond that, further savings would require paying true variational accuracy, which the safeguard forbids.

This connects directly to compilation theory. Exact-placement results [41, 42] and routing lower bounds [44, 46] fix the placement floor; heuristic routers [35, 34, 37] operate on a fixed circuit and cannot go below the exposure written into it. TA-AVQDS acts one level earlier, where the circuit is still plastic. The structural difference between the topologies makes the prediction sharp: on A the exposure is zero by the Hamiltonian-cycle embedding, so there is nothing to gain (observed: multiplier 1.00); on B the exposure is concentrated on the bonds the placement cannot cover, and redirecting growth onto the covered bonds captures most of the routing cost (observed: 22–30% CNOT reduction); on C the exposure is a single periodic bond of weight 1, and the method captures nearly all of it (35–38% reduction).

6.2 Failure analysis: LSM $h_z = -0.7$ on Topology B

On the LSM $h_z = -0.7$ ramp, TA-AVQDS compiles to 240 CNOTs against the naive baseline’s 214, at equal fidelity (0.9990 vs. 0.9991) — the variant is strictly worse, and we report it as such. The logs reveal the mechanism. The LSM pool carries two generators per bond, $X_i X_{i+1}$ and $Y_i Y_{i+1}$,

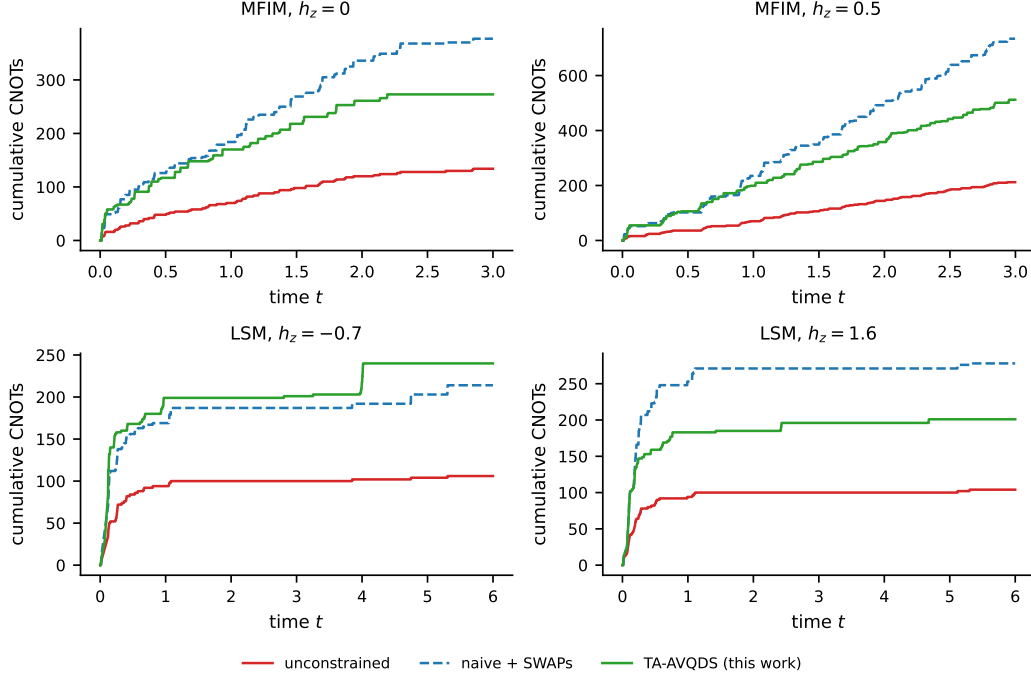


Figure 4: Cumulative CNOT count versus time on Topology B (heavy-hex). The unconstrained reference (red) counts 2 CNOTs per two-qubit rotation under the all-to-all convention; the constrained curves include all SWAPs (3 CNOTs each). TA-AVQDS grows cheaper circuits than naive routing throughout the evolution on three of four benchmarks; on LSM $h_z = -0.7$ it overtakes the naive curve during the post-ramp stage (see Sec. 6.2). The Topology A counterparts are in Fig. 5.

whose trial distances are tied whenever the state is locally x - y -symmetric — which is exactly the situation along the ramp through the $\gamma = 0$ point ($t \approx 1.5$) and during the post-ramp hold at $\gamma = -1$, where the Hamiltonian is pure YY . On those segments the landscape is a web of exact and near-exact degeneracies, and the penalty does not merely reshuffle equivalent choices: it interacts with the secondary $|\dot{\theta}|_{\max}$ tie-break to produce a *larger* ansatz ($N_\theta = 90$ vs. 65) whose extra parameters do buy slightly lower McLachlan distance along the trajectory but not lower final infidelity — the surplus rotations and their routing cost (240 vs. 214 CNOTs at equal SWAP count, 36) are pure overhead. In other words, when the near-degeneracy is exact and persistent rather than incidental, “redirect growth toward cheap generators” can degenerate into “grow more generators,” and the L_{cut}^2 threshold, which normally caps ansatz size, is satisfied either way. The same pathology does not occur at $h_z = 1.6$ because the stronger longitudinal field lifts the x - y symmetry enough that the degeneracies are incidental, and the penalty works as designed (201 vs. 278 CNOTs).

Two honest conclusions follow. First, the failure is bounded and detectable: fidelity is preserved (the safeguard sees to that), and the excess cost is visible in N_θ during the run. Second, the fix is a design question we leave open: penalizing N_θ growth directly, or breaking exact degeneracies by routing cost rather than by $|\dot{\theta}|_{\max}$, would likely recover the naive-baseline cost on this benchmark, but we did not tune the method post hoc to this case. We also note the smaller caveat that on Topology B the variant’s wall-clock cost rises with N_θ (Sec. 6.3), so the failure is also the most expensive run.

6.3 Computational cost

Table 4 reports wall-clock times (single process, 6 BLAS threads). The naive baseline costs the same as unconstrained AVQDS — routing is post-hoc and $\mathcal{O}(\text{gates} \times n)$. TA-AVQDS costs 1.0–1.9 $\times$ the baseline; the increase comes entirely from growing somewhat larger ansätze on Topology B (N_θ : 124 $\rightarrow$ 124, 275 $\rightarrow$ 280, 65 $\rightarrow$ 90, 70 $\rightarrow$ 85), since the trial-expansion loop over the pool dominates the runtime and scales linearly in N_θ . The topology-aware logic itself — one distance lookup

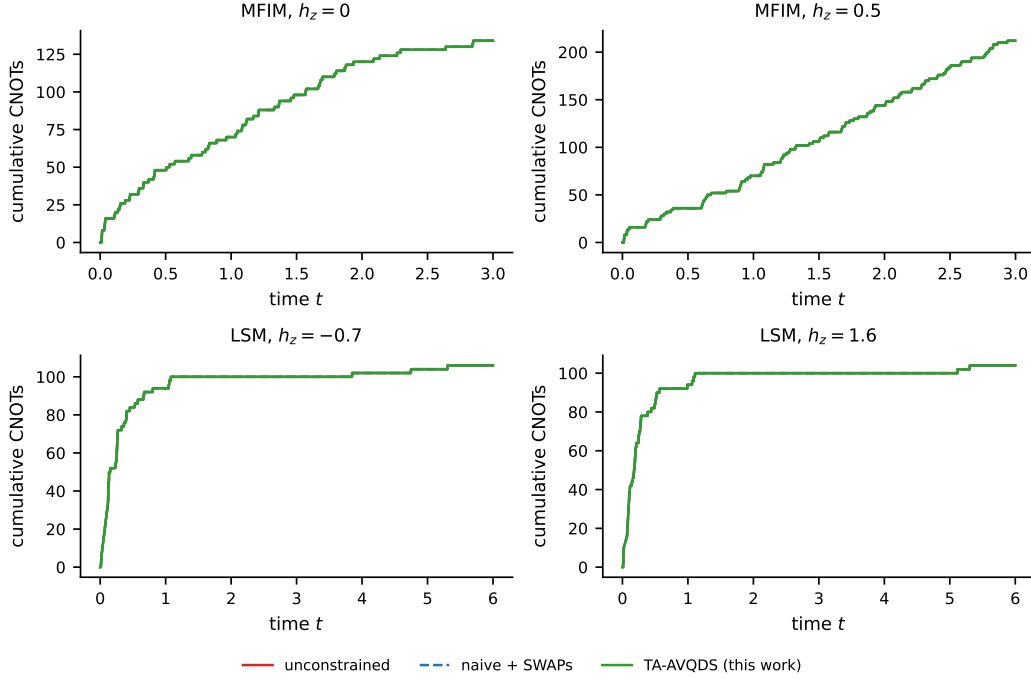


Figure 5: Cumulative CNOT count versus time on Topology A (2×4 square). All three methods coincide with the unconstrained circuit (zero SWAPs, multiplier 1.00), so a single curve is visible per panel.

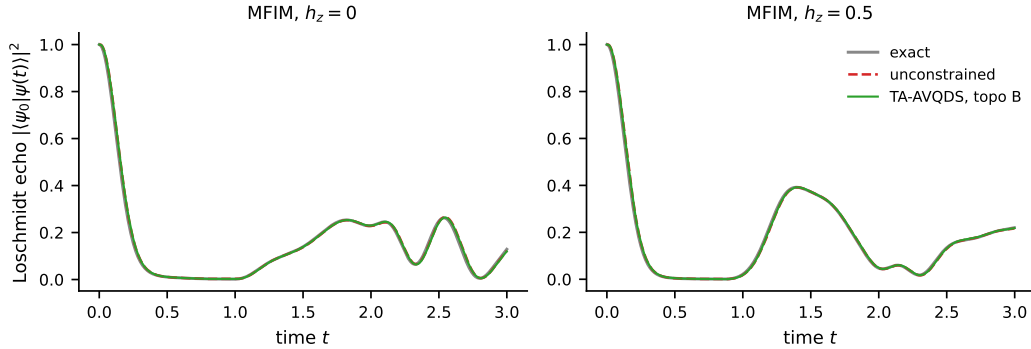


Figure 6: Loschmidt echo $|\langle \Psi_0 | \Psi(t) \rangle|^2$ for the MFIM sudden quenches. The TA-AVQDS circuit compiled to Topology B (green) coincides with the unconstrained circuit (red, dashed) and the exact reference (gray) through the collapse and revival.

per candidate in the selection loop, plus the one-time exact placement (< 1 s) and the greedy routing pass (milliseconds) — is negligible. The absolute times (28–91 s per run) reflect the bit-exact reimplementations on a Pauli-operator backend; the authors’ original dense-matrix code is $\sim 30\times$ slower on the same benchmarks, which is what made the η scans and ablations cheap. No component of the method introduces a superlinear classical cost, and the scaling with system size is the same as AVQDS’s own (exponential statevector memory in exact simulation; on hardware, the trial-distance evaluations replace them).

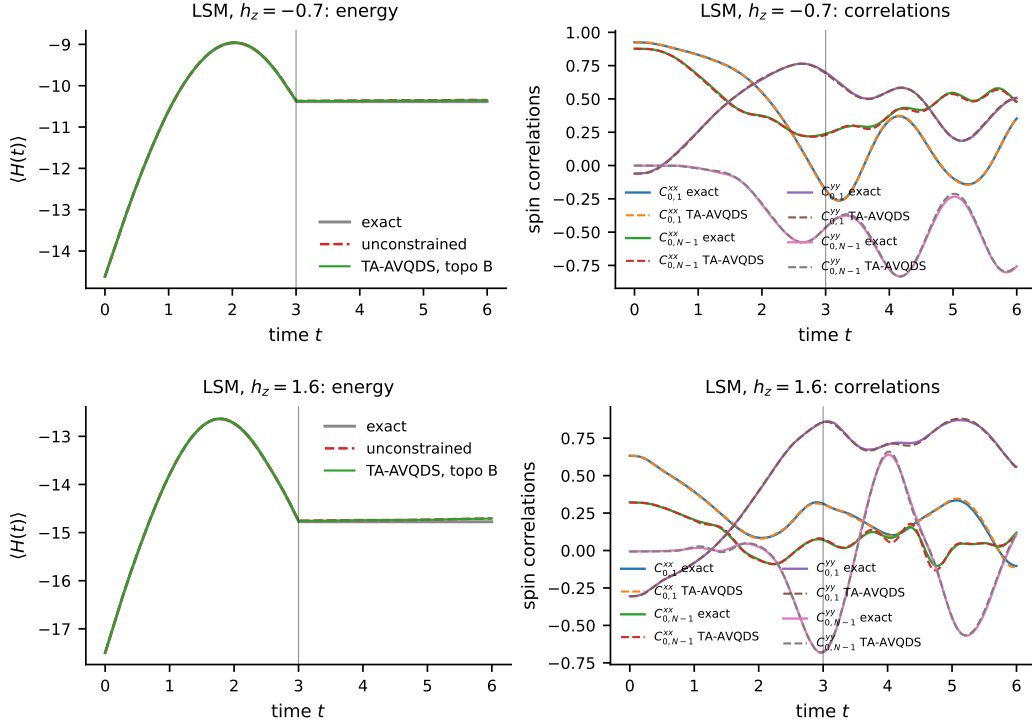


Figure 7: LSM benchmarks (total time $2T = 6$; the ramp ends at $t = 3$, marked by the thin vertical line). Left: energy expectation $\langle \hat{H}(t) \rangle$; right: spin correlations $C_{0,1}^{xx}, C_{0,N-1}^{xx}, C_{0,1}^{yy}, C_{0,N-1}^{yy}$. Solid lines are exact references; dashed curves in the right panels are the TA-AVQDS (Topology B) results, which overlap the exact curves.

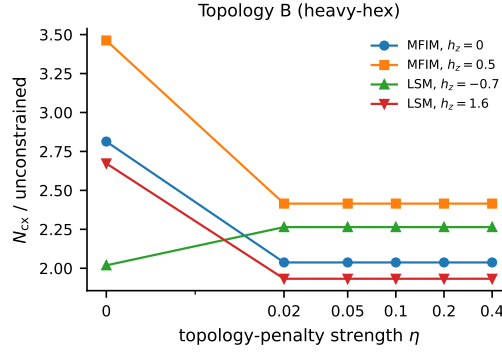


Figure 8: Compiled CNOT multiplier on Topology B versus the penalty strength η (symlog axis). For $\eta = 0$ the method reduces to naive routing; for every $\eta \geq 0.02$ the compiled circuit is identical — the penalty saturates once it exceeds the near-degeneracy scale of the trial-distance landscape. All headline runs use $\eta = 0.1$.

6.4 Third-topology generalization

Nothing in the method is specific to the two required graphs: the placement is an exact solver for any G , the penalty uses only graph distances, and the router is graph-agnostic. Topology C (linear chain) interpolates cleanly between A and B in the exposure hierarchy: LSM embeds perfectly (multiplier 1.00), MFIM exposes exactly one bond, and TA-AVQDS achieves its largest margins there (Table 3). The trend across A, C, B — overhead 1.00, ≈ 2.0 – 2.2 , ≈ 1.9 – 2.4 for the variant,

Table 4: Wall-clock seconds per run (single process, 6 BLAS threads) and final parameter count N_θ . The naive baseline adds negligible post-processing to the unconstrained run; TA-AVQDS’s excess on Topology B tracks its larger ansatz on the affected benchmarks.

	unconstrained	naive A	TA-AVQDS A	naive B	TA-AVQDS B
MFIM $h_z = 0$	37.3	36.2	36.3	36.4	43.2
MFIM $h_z = 0.5$	83.0	83.7	88.0	83.2	91.3
LSM $h_z = -0.7$	27.6	27.8	27.9	27.8	51.3
LSM $h_z = 1.6$	37.8	37.7	37.8	37.8	52.1

versus 1.00, ≈ 3.0 – 3.5 , ≈ 2.0 – 3.5 for naive routing — tracks the fraction of Hamiltonian weight that the placement cannot cover, supporting the exposure-based account of Sec. 6.1.

6.5 Limitations

The study is at $n = 8$ qubits in noiseless statevector simulation; the placement solver is exact but factorial (harmless at $n = 8$, replaced by heuristics such as Refs. [36, 48] at scale), and the interesting question of how the near-degeneracy structure of the trial-distance landscape scales with system size is open. SWAPs are counted but noiseless; on hardware their error would weight the trade-off further toward growth-time avoidance, but would also penalize the residual SWAPs the method still emits — noise, readout error, and error correction on current devices [108, 106, 107, 104, 105] are outside our scope. The LSM $h_z = -0.7$ failure shows the penalty can interact badly with exact pool degeneracies. Finally, we held the AVQDS core fixed by design; combining the penalty with improved integrators [16, 17] or shot-noise-robust selection [18] is future work, as are experimental quench studies of the kind pursued in [101, 102, 103].

7 Conclusion

We have shown that the connectivity constraint of a fixed qubit coupling graph can be folded into the adaptive growth rule of variational quantum dynamics at essentially no cost to the variational principle: an exactly optimized placement removes the constraint entirely on graphs admitting a Hamiltonian cycle, and a safeguarded distance penalty on the trial McLachlan distance redirects growth toward routing-cheap generators where the landscape is near-degenerate, cutting compiled CNOT counts by 22–38% relative to the strongest post-hoc routing baseline on graphs where routing is fundamentally unavoidable. The method’s failure on the x – y -degenerate LSM ramp is equally instructive: growth-time pricing reshuffles degenerate choices, and where degeneracies are exact and persistent, the reshuffling can inflate the ansatz instead of cheapening it. The broader lesson is that adaptive variational algorithms possess a compile-time degree of freedom — the near-degeneracy slack in their selection landscapes — that compilers acting after the fact cannot reach, and that exploiting it requires only a surcharge on the selection objective, not a new ansatz or a new solver.

References

- [1] R. P. Feynman, Simulating physics with computers, *Int. J. Theor. Phys.* **21**, 467 (1982).
- [2] S. Lloyd, Universal quantum simulators, *Science* **273**, 1073 (1996).
- [3] H. F. Trotter, On the product of semi-groups of operators, *Proc. Am. Math. Soc.* **10**, 545 (1959).
- [4] M. Suzuki, Generalized Trotter’s formula and systematic approximants of exponential operators and inner derivations with applications to many-body problems, *Commun. Math. Phys.* **51**, 183 (1976).
- [5] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, Theory of Trotter error with commutator scaling, *Phys. Rev. X* **11**, 011020 (2021).
- [6] A. D. McLachlan, A variational solution of the time-dependent Schrödinger equation, *Mol. Phys.* **8**, 39 (1964).
- [7] Y. Li and S. C. Benjamin, Efficient variational quantum simulator incorporating active error minimization, *Phys. Rev. X* **7**, 021050 (2017).

- [8] X. Yuan, S. Endo, Q. Zhao, Y. Li, and S. C. Benjamin, Theory of variational quantum simulation, *Quantum* **3**, 191 (2019).
- [9] S. Endo, J. Sun, Y. Li, S. C. Benjamin, and X. Yuan, Variational quantum simulation of general processes, *Phys. Rev. Lett.* **125**, 010501 (2020).
- [10] S. Barison, F. Vicentini, and G. Carleo, An efficient quantum algorithm for the time evolution of parameterized circuits, *Quantum* **5**, 512 (2021).
- [11] Y.-X. Yao, N. Gomes, F. Zhang, C.-Z. Wang, K.-M. Ho, T. Iadecola, and P. P. Orth, Adaptive variational quantum dynamics simulations, *PRX Quantum* **2**, 030307 (2021).
- [12] Y.-X. Yao *et al.*, AVQDS reference implementation (avdynamics), accompanying Ref. [11].
- [13] Y.-X. Yao, F. Zhang, C.-Z. Wang, K.-M. Ho, and P. P. Orth, Adaptive variational quantum dynamics simulations of open quantum systems, *arXiv:2003.04211* (2020).
- [14] N. Gomes, A. Mukherjee, F. Zhang, T. Iadecola, C.-Z. Wang, K.-M. Ho, P. P. Orth, and Y.-X. Yao, Adaptive variational quantum imaginary time evolution approach for ground state preparation, *arXiv:2102.01544* (2021).
- [15] F. Zhang, N. Gomes, N. F. Berthussen, P. P. Orth, C.-Z. Wang, K.-M. Ho, and Y.-X. Yao, Shallow-circuit variational quantum eigensolver based on symmetry-protected Hartree-Fock, *Phys. Rev. Research* **3**, 013039 (2021).
- [16] X. Zhang *et al.*, Adaptive variational quantum dynamics with a temporal convolutional integrator, *Phys. Rev. B* **111**, 094310 (2025).
- [17] N. F. Berthussen, T. V. Trevisan, T. Iadecola, and P. P. Orth, Quantum dynamics simulations beyond the coherence time on noisy intermediate-scale quantum hardware by variational Trotter compression, *Phys. Rev. Research* **4**, 023097 (2022).
- [18] H. Zhao, M. Bukov, M. Heyl, and R. Moessner, Making Trotterization adaptive and energy-self-correcting for NISQ devices and beyond, *PRX Quantum* **4**, 030319 (2023).
- [19] H. R. Grimsley, S. E. Economou, E. Barnes, and N. J. Mayhall, An adaptive variational algorithm for exact molecular simulations on a quantum computer, *Nat. Commun.* **10**, 3007 (2019).
- [20] H. L. Tang, V. O. Shkolnikov, G. S. Barron, H. R. Grimsley, N. J. Mayhall, E. Barnes, and S. E. Economou, Qubit-ADAPT-VQE: an adaptive algorithm for constructing hardware-efficient ansätze on a quantum processor, *PRX Quantum* **2**, 020310 (2021).
- [21] Y. S. Yordanov, V. Armaos, C. H. W. Barnes, and D. R. M. Arvidsson-Shukur, Qubit-excitation-based adaptive variational quantum eigensolver, *Commun. Phys.* **4**, 228 (2021).
- [22] D. A. Fedorov, Y. Peng, N. Govind, and Y. Alexeev, VQE method: a short survey and recent developments, *Mater. Theory* **6**, 2 (2022).
- [23] A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik, and J. L. O'Brien, A variational eigenvalue solver on a photonic quantum processor, *Nat. Commun.* **5**, 4213 (2014).
- [24] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles, Variational quantum algorithms, *Nat. Rev. Phys.* **3**, 625 (2021).
- [25] K. Bharti, A. Cervera-Lierta, T. H. Kyaw, T. Haug, S. Alperin-Lea, A. Anand, M. Degroote, H. Heimonen, J. S. Kottmann, T. Menke, W.-K. Mok, S. Sim, L.-C. Kwek, and A. Aspuru-Guzik, Noisy intermediate-scale quantum algorithms, *Rev. Mod. Phys.* **94**, 015004 (2022).
- [26] J. Preskill, Quantum computing in the NISQ era and beyond, *Quantum* **2**, 79 (2018).
- [27] C. Chamberland, G. Zhu, T. J. Yoder, J. B. Hertzberg, and A. W. Cross, Topological and subsystem codes on low-degree graphs with flag qubits, *Phys. Rev. X* **10**, 011022 (2020).
- [28] Y. Kim, A. Eddins, S. Anand, K. X. Wei, E. van den Berg, S. Rosenblatt, H. Nayfeh, Y. Wu, M. Zaletel, K. Temme, and A. Kandala, Evidence for the utility of quantum computing before fault tolerance, *Nature* **618**, 500 (2023).
- [29] F. Arute *et al.*, Quantum supremacy using a programmable superconducting processor, *Nature* **574**, 505 (2019).
- [30] A. Morvan *et al.*, Phase transitions in random circuit sampling, *Nature* **634**, 328 (2024).
- [31] C. D. Bruzewicz, J. Chiaverini, R. McConnell, and J. M. Sage, Trapped-ion quantum computing: progress and challenges, *Appl. Phys. Rev.* **6**, 021314 (2019).
- [32] L. Henriot, L. Beguin, A. Signoles, T. Lahaye, A. Browaeys, G.-O. Reymond, and C. Jurczak, Quantum computing with neutral atoms, *Quantum* **4**, 327 (2020).

- [33] A. Kandala, A. Mezzacapo, K. Temme, M. Takita, M. Brink, J. M. Chow, and J. M. Gambetta, Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets, *Nature* **549**, 242 (2017).
- [34] A. Zulehner, A. Paler, and R. Wille, An efficient methodology for mapping quantum circuits to the IBM QX architectures, *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* **38**, 1226 (2018).
- [35] G. Li, Y. Ding, and Y. Xie, Tackling the qubit mapping problem for NISQ-era quantum devices, in *Proc. ASPLOS*, 1001 (2019).
- [36] S. Sivarajah, S. Dilkes, A. Cowtan, W. Simmons, A. Edgington, and R. Duncan, $t|ket\rangle$: a retargetable compiler for NISQ devices, *Quantum Sci. Technol.* **6**, 014003 (2020).
- [37] A. Cowtan, S. Dilkes, R. Duncan, A. Krajenbrink, W. Simmons, and S. Sivarajah, On the qubit routing problem, in *14th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC)*, LIPIcs **135**, 5:1 (2019).
- [38] R. Wille, L. Burgholzer, and A. Zulehner, Mapping quantum circuits to IBM QX architectures using the minimal number of SWAP and H operations, in *Proc. Design Automation Conference (DAC)*, 142 (2019).
- [39] S. S. Tannu and M. K. Qureshi, Not all qubits are created equal: a case for variability-aware policies for NISQ-era quantum computers, in *Proc. ASPLOS*, 987 (2019).
- [40] P. Murali, J. M. Baker, A. Javadi-Abhari, F. T. Chong, and M. Martonosi, Noise-adaptive compiler mappings for noisy intermediate-scale quantum computers, in *Proc. ASPLOS*, 1015 (2019).
- [41] R. Wille, A. Lye, and R. Drechsler, Exact reordering of circuit lines for nearest neighbor quantum architectures, *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* **33**, 1818 (2014).
- [42] B. Tan and J. Cong, Optimal layout synthesis for quantum computing, in *Proc. ICCAD*, 1 (2020).
- [43] M. Alam, A. Ash-Saki, and S. Ghosh, Circuit compilation methodologies for quantum approximate optimization algorithm, in *Proc. MICRO*, 215 (2020).
- [44] M. Y. Siraichi, V. F. dos Santos, S. Collange, and F. M. Q. Pereira, Qubit allocation, in *Proc. CGO*, 113 (2018).
- [45] A. Botea, A. Kishimoto, and R. Marinescu, On the complexity of quantum circuit compilation, in *Proc. SoCS*, 138 (2018).
- [46] A. M. Childs, E. Schoute, and C. M. Unsal, Circuit transformations for quantum architectures, in *14th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC)*, LIPIcs **135**, 3:1 (2019).
- [47] T. Fösel, M. Y. Niu, F. Marquardt, and L. Li, Quantum circuit optimization with deep reinforcement learning, *arXiv:2103.07585* (2021).
- [48] A. Paler, L. Sasu, A.-C. Florea, and R. Andonie, Machine learning optimization of quantum circuit layouts, *arXiv:2007.10199* (2020).
- [49] D. Wecker, M. B. Hastings, and M. Troyer, Progress towards practical quantum variational algorithms, *Phys. Rev. A* **92**, 042303 (2015).
- [50] I. D. Kivlichan, J. McClean, N. Wiebe, C. Gidney, A. Aspuru-Guzik, G. K.-L. Chan, and R. Babbush, Quantum simulation of electronic structure with linear depth and connectivity, *Phys. Rev. Lett.* **120**, 110501 (2018).
- [51] Z. Jiang, K. J. Sung, K. Kechedzhi, V. N. Smelyanskiy, and S. Boixo, Quantum algorithms to simulate many-body physics of correlated fermions, *Phys. Rev. Applied* **9**, 044036 (2018).
- [52] H. R. Grimsley, G. S. Barron, E. Barnes, S. E. Economou, and N. J. Mayhall, ADAPT-VQE is insensitive to rough parameter landscapes and barren plateaus, *npj Quantum Inf.* **9**, 19 (2023).
- [53] R. Haghshenas, J. Gray, A. C. Potter, and G. K.-L. Chan, Variational power of quantum circuit tensor networks, *Phys. Rev. X* **12**, 011047 (2022).
- [54] A. Martín, L. Laso, D. Pérez-San Pedro, R. Barnes, and A. Encinas, Digital-analog quantum simulations using the cross-resonance effect, *PRX Quantum* **4**, 020321 (2023).
- [55] L. H. Harper, Optimal assignments of numbers to vertices, *J. Soc. Ind. Appl. Math.* **12**, 131 (1964).
- [56] J. Díaz, J. Petit, and M. Serna, A survey of graph layout problems, *ACM Comput. Surv.* **34**, 313 (2002).
- [57] R. Babbush, N. Wiebe, J. McClean, J. McClain, H. Neven, and G. K.-L. Chan, Low-depth quantum simulation of materials, *Phys. Rev. X* **8**, 011044 (2018).

- [58] B. O’Gorman, W. J. Huggins, E. G. Rieffel, and K. B. Whaley, Generalized swap networks for near-term quantum computing, arXiv:1905.05118 (2019).
- [59] R. Barends *et al.*, Digitized adiabatic quantum computing with a superconducting circuit, *Nature* **534**, 222 (2016).
- [60] R. Iten, R. Moyard, T. Metger, D. Sutter, and S. Woerner, Exact and practical pattern matching for quantum circuit optimization, arXiv:1909.05270 (2019).
- [61] D. Venturelli, M. Do, E. Rieffel, and J. Frank, Compiling quantum circuits to realistic hardware architectures using temporal planners, *Quantum Sci. Technol.* **3**, 025004 (2018).
- [62] K. E. C. Booth, M. Do, J. C. Beck, E. Rieffel, D. Venturelli, and J. Frank, Comparing and integrating constraint programming and temporal planning for quantum circuit compilation, in *Proc. ICAPS*, 366 (2018).
- [63] Y. Alexeev *et al.*, Quantum computer systems for scientific discovery, *PRX Quantum* **2**, 017001 (2021).
- [64] E. Lieb, T. Schultz, and D. Mattis, Two soluble models of an antiferromagnetic chain, *Ann. Phys.* **16**, 407 (1961).
- [65] S. Sachdev, *Quantum Phase Transitions*, 2nd ed. (Cambridge University Press, 2011).
- [66] P. Calabrese and J. Cardy, Time dependence of correlation functions following a quantum quench, *Phys. Rev. Lett.* **96**, 136801 (2006).
- [67] J. Eisert, M. Friesdorf, and C. Gogolin, Quantum many-body systems out of equilibrium, *Nat. Phys.* **11**, 124 (2015).
- [68] A. Polkovnikov, K. Sengupta, A. Silva, and M. Vengalattore, Colloquium: Nonequilibrium dynamics of closed interacting quantum systems, *Rev. Mod. Phys.* **83**, 863 (2011).
- [69] J. Haegeman, J. I. Cirac, T. J. Osborne, I. Pizorn, H. Verschelde, and F. Verstraete, Time-dependent variational principle for quantum lattices, *Phys. Rev. Lett.* **107**, 070601 (2011).
- [70] U. Schollwöck, The density-matrix renormalization group in the age of matrix product states, *Ann. Phys.* **326**, 96 (2011).
- [71] G. Vidal, Efficient classical simulation of slightly entangled quantum computations, *Phys. Rev. Lett.* **91**, 147902 (2003).
- [72] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed. (Cambridge University Press, 2010).
- [73] E. Knill, Approximation by quantum circuits, arXiv:quant-ph/9508006 (1995).
- [74] T. Jones and S. C. Benjamin, Robust quantum compilation and circuit optimisation via energy minimisation, *Quantum* **6**, 628 (2022).
- [75] R. S. Smith, M. J. Curtis, and W. J. Zeng, A practical quantum instruction set architecture, arXiv:1608.03355 (2016).
- [76] D. Maslov, S. M. Falconer, and M. Mosca, Quantum circuit simplification and level compaction, *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* **27**, 436 (2008).
- [77] B. Nash, V. Gheorghiu, and M. Mosca, Quantum circuit optimizations for NISQ architectures, *Quantum Sci. Technol.* **5**, 025010 (2020).
- [78] S. Ebadi *et al.*, Quantum phases of matter on a 256-atom programmable quantum simulator, *Nature* **595**, 227 (2021).
- [79] J. R. McClean, J. Romero, R. Babbush, and A. Aspuru-Guzik, The theory of variational hybrid quantum-classical algorithms, *New J. Phys.* **18**, 023023 (2016).
- [80] J. Romero, R. Babbush, J. R. McClean, C. Hempel, P. J. Love, and A. Aspuru-Guzik, Strategies for quantum computing molecular energies using the unitary coupled cluster ansatz, *Quantum Sci. Technol.* **4**, 014008 (2018).
- [81] O. Higgott, D. Wang, and S. Brierley, Variational quantum computation of excited states, *Quantum* **3**, 156 (2019).
- [82] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii, Quantum circuit learning, *Phys. Rev. A* **98**, 032309 (2018).
- [83] M. Schuld, A. Bocharov, K. M. Svore, and N. Wiebe, Circuit-centric quantum classifiers, *Phys. Rev. A* **101**, 032308 (2020).
- [84] M. Lubasch, J. Joo, P. Moinier, M. Kiffner, and D. Jaksch, Variational quantum algorithms for nonlinear problems, *Phys. Rev. A* **101**, 010301(R) (2020).
- [85] C. Cirstoiu, Z. Holmes, J. Iosue, L. Cincio, P. J. Coles, and A. Sornborger, Variational fast forwarding for quantum simulation beyond the coherence time, *npj Quantum Inf.* **6**, 82 (2020).

- [86] J. Gibbs, K. Gili, Z. Holmes, B. Commeau, A. Arrasmith, L. Cincio, P. J. Coles, and A. Sornborger, Long-time simulations with high fidelity on quantum hardware, arXiv:2102.04313 (2021).
- [87] B. Commeau, M. Cerezo, Z. Holmes, L. Cincio, P. J. Coles, and A. Sornborger, Variational Hamiltonian diagonalization for dynamical quantum simulation, arXiv:2009.02559 (2020).
- [88] M. Otten, C. L. Cortes, and S. K. Gray, Noise-resilient quantum dynamics using symmetry-preserving ansatzes, arXiv:1910.06284 (2019).
- [89] W. Lavrijsen, A. Tudor, J. Müller, C. Iancu, and W. de Jong, Classical optimizers for noisy intermediate-scale quantum devices, in *Proc. QCE*, 267 (2020).
- [90] G. G. Guerreschi and M. Smelyanskiy, Practical optimization for hybrid quantum-classical algorithms, arXiv:1701.01450 (2017).
- [91] J. Stokes, J. Izaac, N. Killoran, and G. Carleo, Quantum natural gradient, *Quantum* **4**, 269 (2020).
- [92] B. Koczor and S. C. Benjamin, Quantum analytic descent, *Phys. Rev. Research* **4**, 023017 (2022).
- [93] Y. Du, M.-H. Hsieh, T. Liu, and D. Tao, Expressive power of parametrized quantum circuits, *Phys. Rev. Research* **2**, 033125 (2020).
- [94] Z. Holmes, K. Sharma, M. Cerezo, and P. J. Coles, Connecting ansatz expressibility to gradient magnitudes and barren plateaus, *PRX Quantum* **3**, 010313 (2022).
- [95] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, Barren plateaus in quantum neural network training landscapes, *Nat. Commun.* **9**, 4812 (2018).
- [96] K. Heya, Y. Suzuki, Y. Nakamura, and K. Fujii, Variational quantum gate optimization, arXiv:1810.12745 (2018).
- [97] L. Cincio, Y. Subaşı, A. T. Sornborger, and P. J. Coles, Learning the quantum algorithm for state overlap, *New J. Phys.* **20**, 113022 (2018).
- [98] S. Khatri, R. LaRose, A. Poremba, L. Cincio, A. T. Sornborger, and P. J. Coles, Quantum-assisted quantum compiling, *Quantum* **3**, 140 (2019).
- [99] K. Sharma, S. Khatri, M. Cerezo, and P. J. Coles, Noise resilience of variational quantum compiling, *New J. Phys.* **22**, 043006 (2020).
- [100] M. Heyl, Dynamical quantum phase transitions: a review, *Rep. Prog. Phys.* **81**, 054001 (2018).
- [101] P. Jurcevic, H. Shen, P. Hauke, C. Maier, T. Brydges, C. Hempel, B. P. Lanyon, M. Heyl, R. Blatt, and C. F. Roos, Direct observation of dynamical quantum phase transitions in an interacting many-body system, *Phys. Rev. Lett.* **119**, 080501 (2017).
- [102] J. Zhang, G. Pagano, P. W. Hess, A. Kyprianidis, P. Becker, H. Kaplan, A. V. Gorshkov, Z.-X. Gong, and C. Monroe, Observation of a many-body dynamical phase transition with a 53-qubit quantum simulator, *Nature* **551**, 601 (2017).
- [103] W. L. Tan, P. Becker, F. Liu, G. Pagano, K. S. Collins, A. De, L. Feng, H. B. Kaplan, A. Kyprianidis, R. Lundgren, W. Morong, S. Whitsitt, A. V. Gorshkov, and C. Monroe, Domain-wall confinement and dynamics in a quantum simulator, *Nat. Phys.* **17**, 742 (2021).
- [104] X. Mi *et al.*, Time-crystalline eigenstate order on a quantum processor, *Nature* **601**, 531 (2022).
- [105] P. Frey and S. Rachel, Realization of a discrete time crystal on 57 qubits of a quantum computer, *Sci. Adv.* **8**, eabm7652 (2022).
- [106] Google Quantum AI and Collaborators, Measurement-induced entanglement and quantum computation on a noisy superconducting processor, *Nature* **622**, 481 (2023).
- [107] D. Bluvstein *et al.*, Logical quantum processor based on reconfigurable atom arrays, *Nature* **626**, 58 (2024).
- [108] R. Acharya *et al.*, Quantum error correction below the surface code threshold, *Nature* **638**, 920 (2025).